

Seat No: .....

Enrollment No: .....

**PARUL UNIVERSITY**  
**FACULTY OF IT & COMPUTER SCIENCE**  
**BCA/BCA(Hons)/B.Sc.(IT)/B.Sc.(IT) (Hons), Summer 2024-25 Examination**

Semester: IV

Subject Code: 05010104DS07/05010204DS07/05101266

Subject Name: Big data Analytics-I (T)/Introduction to  
Big data Analytics(T)/Big Data Analytics – I

Date: 22-04-2025

Time: 02:00 pm to 04:30 pm

Total Marks: 60

**Instructions:**

1. All questions are compulsory.
2. Make suitable assumptions whenever necessary.
3. Write the answers for both sections on separate answer sheets.

| SECTION - A [30 Marks] |                                                                                                                                                                                                                                                                                                                       |      |     |     |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-----|-----|
| Q1                     | Answer the following questions:                                                                                                                                                                                                                                                                                       | [20] | CO  | BT  |
| (a)                    | Answer <b>all</b> questions                                                                                                                                                                                                                                                                                           | (8)  |     |     |
| i.                     | A company previously used Excel spreadsheets for data storage but now uses Hadoop. What advantage does this transition offer?<br>(A).Increased manual data entry (B).Slower processing of large datasets (C).Ability to process and analyze massive datasets efficiently (D).Less dependency on distributed computing | 1    | CO2 | BT1 |
| ii.                    | Custom extensions built in the ____ programming language are also supported by Hive.<br>(A) Java (B) C# (C) C (D) C++                                                                                                                                                                                                 | 1    | CO3 | BT1 |
| iii.                   | What is a key source of Big Data in detecting fraud?                                                                                                                                                                                                                                                                  | 1    | CO1 | BT2 |
| iv.                    | What is unstructured data ?                                                                                                                                                                                                                                                                                           | 1    | CO1 | BT1 |
| v.                     | What is a key feature of Apache Ignite in in-memory computing?                                                                                                                                                                                                                                                        | 1    | CO2 | BT2 |
| vi.                    | What is the function of HDFS in Hadoop?                                                                                                                                                                                                                                                                               | 1    | CO2 | BT3 |
| vii.                   | In Spark, which component is responsible for in-memory computation?                                                                                                                                                                                                                                                   | 1    | CO2 | BT1 |
| viii.                  | What is the role of NameNode in Hadoop?                                                                                                                                                                                                                                                                               | 1    | CO2 | BT2 |
| (b)                    | Attempt Any <b>Four</b> Questions out of Six                                                                                                                                                                                                                                                                          | (12) |     |     |
| i.                     | Compare Batch Processing and Real-time Processing in Big Data.                                                                                                                                                                                                                                                        | 3    | CO1 | BT4 |
| ii.                    | Discuss the advantages of Big Data analytics for businesses.                                                                                                                                                                                                                                                          | 3    | CO1 | BT3 |

|                               |                                                                                                                                                                                                                                                                                |             |           |           |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-----------|-----------|
| iii.                          | Discuss a real-world application of parallel computing in Big Data analytics.                                                                                                                                                                                                  | 3           | CO2       | BT3       |
| iv.                           | How can Big Data analytics prevent insurance fraud?                                                                                                                                                                                                                            | 3           | CO1       | BT3       |
| v.                            | How does SequenceFileInputFormat differ from TextInputFormat?                                                                                                                                                                                                                  | 3           | CO3       | BT4       |
| vi.                           | Explain the advantages of Hadoop over traditional databases.                                                                                                                                                                                                                   | 3           | CO2       | BT4       |
|                               |                                                                                                                                                                                                                                                                                |             |           |           |
| <b>Q2</b>                     | <b>Answer/Solve following in detail</b>                                                                                                                                                                                                                                        | <b>[10]</b> |           |           |
| (a)                           | Describe a situation where the transition from traditional databases to Big Data was beneficial.                                                                                                                                                                               | 5           | CO1       | BT5       |
| (b)                           | What is the purpose of the -put command in Hadoop CLI, and how is it used?                                                                                                                                                                                                     | 5           | CO3       | BT4       |
| <b>OR</b>                     |                                                                                                                                                                                                                                                                                |             |           |           |
| (a)                           | Design an optimal Big Data architecture to handle diverse data sources.                                                                                                                                                                                                        | 5           | CO1       | BT6       |
| (b)                           | How does In-Memory Computing impact the performance of real-time analytics in Big Data systems?                                                                                                                                                                                | 5           | CO2       | BT4       |
|                               |                                                                                                                                                                                                                                                                                |             |           |           |
| <b>SECTION - B [30 Marks]</b> |                                                                                                                                                                                                                                                                                |             |           |           |
| <b>Q1</b>                     | <b>Answer the following questions:</b>                                                                                                                                                                                                                                         | <b>[20]</b> | <b>CO</b> | <b>BT</b> |
| (a)                           | Answer <b>all</b> questions                                                                                                                                                                                                                                                    | (8)         |           |           |
| i.                            | Amongst which of the following is / are correct,<br>(A).Hive is a relational database that supports SQL queries. (B).Pig is a relational database that supports SQL queries. (C).Both A and B (D).None of the mentioned above                                                  | 1           | CO3       | BT1       |
| ii.                           | What is the primary role of Big Data in insurance fraud detection?<br>(A).Increasing the cost of insurance claims (B).Identifying suspicious patterns and anomalies in claims (C).Approving all insurance claims automatically (D).Preventing customers from submitting claims | 1           | CO1       | BT1       |
| iii.                          | What is the main advantage of using distributed computing in Big Data processing?                                                                                                                                                                                              | 1           | CO2       | BT2       |
| iv.                           | Which Big Data technology is commonly used for processing large-scale social media data?                                                                                                                                                                                       | 1           | CO1       | BT3       |
| v.                            | What is the default replication factor in HDFS?                                                                                                                                                                                                                                | 1           | CO2       | BT1       |
| vi.                           | How does Big Sheet process data?                                                                                                                                                                                                                                               | 1           | CO2       | BT2       |
| vii.                          | How do you specify the Mapper and Reducer in a Hadoop Streaming job?                                                                                                                                                                                                           | 1           | CO2       | BT3       |
| viii.                         | Which tool in the Hadoop ecosystem is used for querying large datasets with SQL-like syntax?                                                                                                                                                                                   | 1           | CO2       | BT2       |
|                               |                                                                                                                                                                                                                                                                                |             |           |           |
| (b)                           | Attempt Any <b>Four</b> Questions out of Six                                                                                                                                                                                                                                   | (12)        |           |           |

|           |                                                                                                            |             |     |     |
|-----------|------------------------------------------------------------------------------------------------------------|-------------|-----|-----|
| i.        | Explain the role of predictive analytics in retail using Big Data.                                         | 3           | CO1 | BT2 |
| ii.       | Compare on-premise Big Data solutions with cloud-based Big Data solutions.                                 | 3           | CO2 | BT4 |
| iii.      | What is Big Data in Social Networking?                                                                     | 3           | CO1 | BT2 |
| iv.       | Describe a scenario where Big Sheet can be useful.                                                         | 3           | CO3 | BT3 |
| v.        | How can you handle large-scale log processing using Hadoop Streaming?                                      | 3           | CO3 | BT3 |
| vi.       | How does a client read a file in HDFS?                                                                     | 3           | CO3 | BT3 |
|           |                                                                                                            |             |     |     |
| <b>Q2</b> | <b>Answer/Solve following in detail</b>                                                                    | <b>[10]</b> |     |     |
| (a)       | How do external data sources, such as social media, contribute to detecting fraud in the insurance sector? | 5           | CO1 | BT5 |
| (b)       | Explain the core components of InfoSphere BigInsights.                                                     | 5           | CO2 | BT4 |
| <b>OR</b> |                                                                                                            |             |     |     |
| (a)       | What happens if there is a failure in the shuffle and sort phase?                                          | 5           | CO4 | BT4 |
| (b)       | What is MapReduce, and how does it fit into the Hadoop Ecosystem?                                          | 5           | CO2 | BT4 |
|           |                                                                                                            |             |     |     |

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**BCA/BCA(Hons)/B.Sc.(IT)/B.Sc.(IT) (Hons), Summer 2024-25 Examination**

Semester: IV

Subject Code: 05010104DS07/05010204DS07/05101266

Subject Name: Big data Analytics-I (T)/Introduction to  
Big data Analytics(T)/Big Data Analytics – I

Date: 22-04-2025

Time: 02:00 pm to 04:30 pm

Total Marks: 60

**Instructions:**

1. All questions are compulsory.
2. Make suitable assumptions whenever necessary.
3. Write the answers for both sections on separate answer sheets.

| SECTION - A [30 Marks] |                                                                                                                                                                                                                                                                                                                       |      |     |     |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-----|-----|
| Q1                     | Answer the following questions:                                                                                                                                                                                                                                                                                       | [20] | CO  | BT  |
| (a)                    | Answer <b>all</b> questions                                                                                                                                                                                                                                                                                           | (8)  |     |     |
| i.                     | A company previously used Excel spreadsheets for data storage but now uses Hadoop. What advantage does this transition offer?<br>(A).Increased manual data entry (B).Slower processing of large datasets (C).Ability to process and analyze massive datasets efficiently (D).Less dependency on distributed computing | 1    | CO2 | BT1 |
| ii.                    | Custom extensions built in the ____ programming language are also supported by Hive.<br>(A) Java (B) C# (C) C (D) C++                                                                                                                                                                                                 | 1    | CO3 | BT1 |
| iii.                   | What is a key source of Big Data in detecting fraud?                                                                                                                                                                                                                                                                  | 1    | CO1 | BT2 |
| iv.                    | What is unstructured data ?                                                                                                                                                                                                                                                                                           | 1    | CO1 | BT1 |
| v.                     | What is a key feature of Apache Ignite in in-memory computing?                                                                                                                                                                                                                                                        | 1    | CO2 | BT2 |
| vi.                    | What is the function of HDFS in Hadoop?                                                                                                                                                                                                                                                                               | 1    | CO2 | BT3 |
| vii.                   | In Spark, which component is responsible for in-memory computation?                                                                                                                                                                                                                                                   | 1    | CO2 | BT1 |
| viii.                  | What is the role of NameNode in Hadoop?                                                                                                                                                                                                                                                                               | 1    | CO2 | BT2 |
| (b)                    | Attempt Any <b>Four</b> Questions out of Six                                                                                                                                                                                                                                                                          | (12) |     |     |
| i.                     | Compare Batch Processing and Real-time Processing in Big Data.                                                                                                                                                                                                                                                        | 3    | CO1 | BT4 |
| ii.                    | Discuss the advantages of Big Data analytics for businesses.                                                                                                                                                                                                                                                          | 3    | CO1 | BT3 |

|                               |                                                                                                                                                                                                                                                                                |             |           |           |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-----------|-----------|
| iii.                          | Discuss a real-world application of parallel computing in Big Data analytics.                                                                                                                                                                                                  | 3           | CO2       | BT3       |
| iv.                           | How can Big Data analytics prevent insurance fraud?                                                                                                                                                                                                                            | 3           | CO1       | BT3       |
| v.                            | How does SequenceFileInputFormat differ from TextInputFormat?                                                                                                                                                                                                                  | 3           | CO3       | BT4       |
| vi.                           | Explain the advantages of Hadoop over traditional databases.                                                                                                                                                                                                                   | 3           | CO2       | BT4       |
|                               |                                                                                                                                                                                                                                                                                |             |           |           |
| <b>Q2</b>                     | <b>Answer/Solve following in detail</b>                                                                                                                                                                                                                                        | <b>[10]</b> |           |           |
| (a)                           | Describe a situation where the transition from traditional databases to Big Data was beneficial.                                                                                                                                                                               | 5           | CO1       | BT5       |
| (b)                           | What is the purpose of the -put command in Hadoop CLI, and how is it used?                                                                                                                                                                                                     | 5           | CO3       | BT4       |
| <b>OR</b>                     |                                                                                                                                                                                                                                                                                |             |           |           |
| (a)                           | Design an optimal Big Data architecture to handle diverse data sources.                                                                                                                                                                                                        | 5           | CO1       | BT6       |
| (b)                           | How does In-Memory Computing impact the performance of real-time analytics in Big Data systems?                                                                                                                                                                                | 5           | CO2       | BT4       |
|                               |                                                                                                                                                                                                                                                                                |             |           |           |
| <b>SECTION - B [30 Marks]</b> |                                                                                                                                                                                                                                                                                |             |           |           |
| <b>Q1</b>                     | <b>Answer the following questions:</b>                                                                                                                                                                                                                                         | <b>[20]</b> | <b>CO</b> | <b>BT</b> |
| (a)                           | Answer <b>all</b> questions                                                                                                                                                                                                                                                    | (8)         |           |           |
| i.                            | Amongst which of the following is / are correct,<br>(A).Hive is a relational database that supports SQL queries. (B).Pig is a relational database that supports SQL queries. (C).Both A and B (D).None of the mentioned above                                                  | 1           | CO3       | BT1       |
| ii.                           | What is the primary role of Big Data in insurance fraud detection?<br>(A).Increasing the cost of insurance claims (B).Identifying suspicious patterns and anomalies in claims (C).Approving all insurance claims automatically (D).Preventing customers from submitting claims | 1           | CO1       | BT1       |
| iii.                          | What is the main advantage of using distributed computing in Big Data processing?                                                                                                                                                                                              | 1           | CO2       | BT2       |
| iv.                           | Which Big Data technology is commonly used for processing large-scale social media data?                                                                                                                                                                                       | 1           | CO1       | BT3       |
| v.                            | What is the default replication factor in HDFS?                                                                                                                                                                                                                                | 1           | CO2       | BT1       |
| vi.                           | How does Big Sheet process data?                                                                                                                                                                                                                                               | 1           | CO2       | BT2       |
| vii.                          | How do you specify the Mapper and Reducer in a Hadoop Streaming job?                                                                                                                                                                                                           | 1           | CO2       | BT3       |
| viii.                         | Which tool in the Hadoop ecosystem is used for querying large datasets with SQL-like syntax?                                                                                                                                                                                   | 1           | CO2       | BT2       |
|                               |                                                                                                                                                                                                                                                                                |             |           |           |
| (b)                           | Attempt Any <b>Four</b> Questions out of Six                                                                                                                                                                                                                                   | (12)        |           |           |

|           |                                                                                                            |             |     |     |
|-----------|------------------------------------------------------------------------------------------------------------|-------------|-----|-----|
| i.        | Explain the role of predictive analytics in retail using Big Data.                                         | 3           | CO1 | BT2 |
| ii.       | Compare on-premise Big Data solutions with cloud-based Big Data solutions.                                 | 3           | CO2 | BT4 |
| iii.      | What is Big Data in Social Networking?                                                                     | 3           | CO1 | BT2 |
| iv.       | Describe a scenario where Big Sheet can be useful.                                                         | 3           | CO3 | BT3 |
| v.        | How can you handle large-scale log processing using Hadoop Streaming?                                      | 3           | CO3 | BT3 |
| vi.       | How does a client read a file in HDFS?                                                                     | 3           | CO3 | BT3 |
|           |                                                                                                            |             |     |     |
| <b>Q2</b> | <b>Answer/Solve following in detail</b>                                                                    | <b>[10]</b> |     |     |
| (a)       | How do external data sources, such as social media, contribute to detecting fraud in the insurance sector? | 5           | CO1 | BT5 |
| (b)       | Explain the core components of InfoSphere BigInsights.                                                     | 5           | CO2 | BT4 |
| <b>OR</b> |                                                                                                            |             |     |     |
| (a)       | What happens if there is a failure in the shuffle and sort phase?                                          | 5           | CO4 | BT4 |
| (b)       | What is MapReduce, and how does it fit into the Hadoop Ecosystem?                                          | 5           | CO2 | BT4 |
|           |                                                                                                            |             |     |     |

**PARUL UNIVERSITY**  
**FACULTY OF ENGINEERING & TECHNOLOGY**  
**B.Tech. Winter Examination 2023 - 24**

**Semester: 6/7**  
**Subject Code: 203105443**  
**Subject Name: Big Data Analytics**

**Date: 01/11/2023**  
**Time: 10:30 am to 01:00 pm**  
**Total Marks: 60**

**Instructions:**

1. All questions are compulsory.
2. Figures to the right indicate full marks.
3. Make suitable assumptions wherever necessary.
4. Start new question on new page.

**Q.1 Objective Type Questions****(15)**

1. Which of the following is NOT a primary characteristic of Big Data?
  - A. Volume
  - B. Variety
  - C. Velocity
  - D. Veracity
2. What does the term "ETL" stand for in the context of Big Data Analytics?
  - A. Extract, Transform, Load
  - B. Efficient Textual Logging
  - C. Event Tracking Language
  - D. Easy Table Lookup
3. What is the goal of A/B testing in the context of Big Data Analytics?
  - A. To analyze data in alphabetical order
  - B. To compare two versions of a web page or app to determine which one performs better
  - C. To store data in a distributed file system
  - D. To conduct data encryption for security purposes
4. In Big Data Analytics, what does the term "MapReduce" refer to?
  - A. A data visualization technique
  - B. A data preprocessing step
  - C. A programming model for parallel processing
  - D. A type of data compression algorithm
5. What is the primary purpose of a data warehouse in the context of Big Data Analytics?
  - A. To store raw data in its original form
  - B. To archive historical data for long-term storage
  - C. To provide a central repository for structured data used for reporting and analysis
  - D. To host data lakes for unstructured data storage
6. Which programming language is commonly used for data analysis in Big Data Analytics?
  - A. Java
  - B. Python
  - C. C++
  - D. Ruby
7. Which of the following is a common technique for handling missing data in Big Data Analytics?
  - A. Deleting all rows with missing data
  - B. Replacing missing values with zeros
  - C. Imputing missing values with statistical measures like mean or median
  - D. Ignoring missing data and proceeding with the analysis
8. What is the primary goal of data compression in Big Data Analytics?
  - A. To increase data storage costs
  - B. To reduce the speed of data access

- C. To save storage space and improve data transfer efficiency
  - D. To increase data security
9. What does the acronym OLAP stand for in the context of Big Data Analytics?
- A. Online Learning and Prediction
  - B. On-Demand Linear Analysis Platform
  - C. Online Analytical Processing
  - D. Outlier Location and Analysis Protocol
10. Which of the following is a key advantage of in-memory processing in Big Data Analytics?
- A. It reduces data storage costs.
  - B. It allows for real-time data access and analysis.
  - C. It increases data variety.
  - D. It requires less computational power.
11. Which of the following is a key component of the Apache Hadoop ecosystem for processing large datasets?
- A. Apache Spark
  - B. Apache Kafka
  - C. Apache Hive
  - D. Apache Tomcat
12. Which of the following is NOT a common data visualization tool used in Big Data Analytics?
- A. Tableau
  - B. Matplotlib
  - C. Power BI
  - D. Hadoop
13. In the context of Big Data Analytics, what does the term "data aggregation" refer to?
- A. The process of cleaning and preparing data
  - B. Combining data from multiple sources into a single dataset
  - C. Storing data in a distributed file system
  - D. Encrypting sensitive data for privacy protection
14. Which technology is commonly used for real-time stream processing in Big Data Analytics?
- A. Hadoop
  - B. Spark
  - C. Flink
  - D. Cassandra
15. In Big Data Analytics, what is the term "data munging" or "data wrangling" referring to?
- A. The process of data compression
  - B. The process of data cleansing and transformation
  - C. The process of data encryption
  - D. The process of data sampling

**Q.2 Answer the following questions. (Attempt any three)**

**(15)**

- A) Explain the key components of the Hadoop ecosystem and how they work together to process and analyze large datasets. Provide examples of when and why Hadoop is a suitable solution for big data processing.
- B) Explain the key characteristics and advantages of NoSQL databases over traditional relational databases. Provide examples of common types of NoSQL databases and their typical use cases.



- C) What are the core component of Hadoop's MapReduce paradigm?  
D) Write a short note on Hadoop Architecture.
- Q.3** A) How does scalability differ between Traditional BI and Big Data, and why is it important? **(07)**  
B) Write a short note on Jasper Reports **(08)**
- OR**
- B) Explain MongoDB with its features that make it better than MySQL? **(08)**
- Q.4** A) What is big data? Where does it come from? How does it work? **(07)**
- OR**
- A) Elaborate the difference between Pig and Hive in details. **(07)**  
B) Explain the CAP theorem and its implications for NoSQL databases. **(08)**  
How do NoSQL databases typically trade off between consistency, availability, and partition tolerance?

**PARUL UNIVERSITY**  
**FACULTY OF ENGINEERING & TECHNOLOGY**  
**B.Tech. Summer 2023 - 24 Examination**

**Semester: 6 & 7**  
**Subject Code: 203105443**  
**Subject Name: Big Data Analytics**

**Date: 29/04/2024**  
**Time: 10.30 am to 1.00 pm**  
**Total Marks: 60**

**Instructions:**

1. All questions are compulsory.
2. Figures to the right indicate full marks.
3. Make suitable assumptions wherever necessary.
4. Start new question on new page.

| <b>Q.1 Objective Type Questions</b> —(All are compulsory) (Each of one mark)                                                                                                                                                                           | <b>(15)</b> | <b>CO</b> | <b>PO</b> | <b>Bloom's Taxonomy</b> |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-----------|-----------|-------------------------|
| 1. What defines Big Data?<br>a) Data that is large in size<br>b) Data that cannot be processed<br>c) Data that exceeds the capacity of traditional databases<br>d) Data that is unstructured                                                           | <b>(01)</b> | <b>1</b>  | <b>1</b>  | <b>Understand</b>       |
| 2. Which of the following is not one of the three Vs used to characterize Big Data?<br>a) Volume<br>b) Variety<br>c) Velocity<br>d) Validation                                                                                                         | <b>(01)</b> | <b>2</b>  | <b>1</b>  | <b>Understand</b>       |
| 3. Which industry often uses analytics for fraud detection and risk assessment?<br>a) Healthcare<br>b) Retail<br>c) Finance<br>d) Education                                                                                                            | <b>(01)</b> | <b>2</b>  | <b>5</b>  | <b>Remember</b>         |
| 4. What is the primary goal of initial data exploration and analysis?<br>a) To build complex machine learning models<br>b) To generate detailed reports<br>c) To gain a preliminary understanding of the data<br>d) To create data visualizations only | <b>(01)</b> | <b>1</b>  | <b>2</b>  | <b>Analyze</b>          |
| 5. Which of the following is NOT a common step in initial data exploration?<br>a) Data cleaning and preprocessing<br>b) Creating advanced machine learning models<br>c) Descriptive statistics<br>d) Data visualization                                | <b>(01)</b> | <b>1</b>  | <b>2</b>  | <b>Apply</b>            |
| 6. What is the term for solving analytics problems within a database using tools like MADlib?                                                                                                                                                          | <b>(01)</b> | <b>5</b>  | <b>5</b>  | <b>Evaluate</b>         |
| 7. What is the term for the final phase of an analytics project where insights are put into practical use within an organization?                                                                                                                      | <b>(01)</b> | <b>4</b>  | <b>4</b>  | <b>Analyze</b>          |
| 8. What is the word for the process of examining and verifying the accuracy and validity of data, models, or results?                                                                                                                                  | <b>(01)</b> | <b>4</b>  | <b>2</b>  | <b>Remember</b>         |
| 9. What is the single word for the systematic approach of fitting models, scoring data, and evaluating their performance in analytics?                                                                                                                 | <b>(01)</b> | <b>3</b>  | <b>8</b>  | <b>Understand</b>       |
| 10. What is the word for the initial assumptions or educated guesses made before conducting an analysis or experiment?                                                                                                                                 | <b>(01)</b> | <b>1</b>  | <b>12</b> | <b>Remember</b>         |

|            |                                                                                                                                                                                              |      |   |   |            |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|---|---|------------|
|            | 11. To perform analytics on unstructured data using the MapReduce programming paradigm, organizations can use _____.                                                                         | (01) | 2 | 4 | Apply      |
|            | 12. In the Hadoop ecosystem, _____ is commonly used for distributed data storage and processing.                                                                                             | (01) | 4 | 7 | Analyze    |
|            | 13. Organizations can leverage _____ for querying and processing data in a Hadoop environment.                                                                                               | (01) | 4 | 2 | Evaluate   |
|            | 14. In Advanced Analytics and Statistical Modeling for Big Data, it is essential to select an appropriate technique based on _____, initial hypotheses, and the data's structure and volume. | (01) | 3 | 1 | Apply      |
|            | 15. Understanding the specific _____ is crucial to determine where each analytical technique can provide the most value.                                                                     | (01) | 2 | 1 | Understand |
| <b>Q.2</b> | <b>Answer the following questions. (Attempt any three)</b>                                                                                                                                   | (15) |   |   |            |
|            | A) How has the emergence of Big Data transformed traditional data storage and processing methods? Provide key examples of Big Data technologies.                                             | (05) | 1 | 7 | Understand |
|            | B) Discuss the three Vs of Big Data (Volume, Velocity, Variety) and how they pose challenges and opportunities for data analysis.                                                            | (05) | 2 | 6 | Understand |
|            | C) Explain the importance of initial exploration and analysis when working with a new dataset. What are the primary objectives of this phase?                                                | (05) | 1 | 3 | Apply      |
|            | D) Discuss the process of applying statistical tests to assess the significance of observed differences or relationships in data.                                                            | (05) | 3 | 5 | Analyze    |
| <b>Q.3</b> | A) Provide a detailed overview of the Data Analytics Lifecycle and its key stages. Why is each stage essential for successful data analysis?                                                 | (07) | 3 | 5 | Evaluate   |
|            | B) How can data visualization aid in the initial exploration and analysis of data? Provide examples of different types of data visualizations that are useful at this stage.                 | (08) | 4 | 4 | Create     |
|            | <b>OR</b>                                                                                                                                                                                    |      |   |   |            |
|            | B) Discuss the challenges and strategies involved in the "Data Preparation" stage, including data cleaning, transformation, and integration.                                                 | (08) | 5 | 2 | Evaluate   |
| <b>Q.4</b> | A) Discuss the advantages of using histograms and box plots as data visualization techniques to understand data distributions.                                                               | (07) | 3 | 3 | Evaluate   |
|            | <b>OR</b>                                                                                                                                                                                    |      |   |   |            |
|            | A) Describe how the retail and e-commerce sectors utilize Big Data analytics to enhance customer experiences, optimize pricing strategies, and improve inventory management.                 | (07) | 4 | 3 | Apply      |
|            | B) Explore the applications of Big Data analytics in the healthcare industry, including patient care, disease diagnosis, and pharmaceutical research.                                        | (08) | 5 | 4 | Analyze    |

Seat No: \_\_\_\_\_

Enrollment No: \_\_\_\_\_

**PARUL UNIVERSITY**  
**FACULTY OF ENGINEERING & TECHNOLOGY**  
**M.Tech., Summer 2017-18 Examination**

**Semester: 2**

**Subject Code: 03201152**

**Subject Name: Big Data Analytics**

**Date: 21/05/2018**

**Time: 2:00 pm to 4:30 pm**

**Total Marks: 60**

**Instructions:**

1. All questions are compulsory.
2. Figures to the right indicate full marks.
3. Make suitable assumptions wherever necessary.
4. Start new question on new page.

**Q.1** A) Explain for 4 v's of Big Data. Write any two characteristics of Big Data. **(05)**

B) What do you mean by HiveQL Data Definition Language? Explain any three HiveQL **(05)**

DDL command with its syntax and example.

C) Why we use MongoDB and compare MongoDB with other databases. **(05)**

**Q.2** Answer the following questions. (Attempt any three) (Each five mark) **(15)**

A) Write short note on Apache Pig. Compare Apache Pig with Map Reduce.

B) List any 5 commands of Hadoop file System. Explain each briefly.

C) Write a short note on NoSQL database. Differentiate NoSQL and relational Database.

D) Explain YARN Architecture.

**Q.3** A). Explain Job Scheduling in Map Reduce. How it is done in case of (i) The Fair Scheduler (ii) The Capacity Scheduler. **(07)**

B). Explain high level Hadoop Architecture. Also Explain following component of it: **(08)**

a) Name Node

b) Data Node

c) Task tracker

d) Job tracker

**OR**

B). Explain HIVE architecture and types of data models of HIVE. **(08)**

**Q.4** A). Write a program for Map Reduce to count occurrences of specific words in the input textfile(s). Also write the commands to compile and run of this code. **(07)**

**OR**

A). Discuss challenges under Big Data. How Big Data Analytics can be useful in The development of smart cities. **(07)**

B). Explain working of following phases of Map Reduce with one common example. **(08)**

a) Map Phase

b) Combiner Phase

c) Shuffle and Sort Phase

d) Reducer Phase.

**PARUL UNIVERSITY**  
**FACULTY OF ENGINEERING & TECHNOLOGY**  
**M.Tech. Winter 2017 - 18 Examination**

**Semester: 2**  
**Subject Code: 03201152**  
**Subject Name: Big Data Analytics**

**Date: 09/01/2018**  
**Time: 02:00pm to 04:30pm**  
**Total Marks: 60**

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**Instructions:**

1. All questions are compulsory.
2. Figures to the right indicate full marks.
3. Make suitable assumptions wherever necessary.
4. Start new question on new page.

**Q.1A) What are Structured, semi-structured and unstructured data in Big Data? (05)**

B) Explain various challenges of Big Data Analytics. (05)

C) Explain the 4V's of Big Data with example. (05)

**Q.2 Answer the following questions. (Attempt any three) (Each five mark) (15)**

A) Explain how Big Data can be deployed in E-Commerce?

B) Differentiate traditional Business Intelligence and Big Data Approach.

C) Differentiate RDBMS and HADOOP.

D) Differentiate SQL, NoSQL and New SQL.

**Q.3A) What is Map Reduce? Explain Map Reduce Work Flow. (07)**

B) Explain MongoDB with its features that make it better than MySQL? (08)

**OR**

B) Write short note on Hadoop Architecture. (08)

**Q.4A) What is Hive? Explain HiveQL with Example. (07)**

**OR**

A) Draw & Explain HDFS Architecture. (07)

B) Write short note on Jasper Reports. (08)

**PARUL UNIVERSITY**  
**FACULTY OF ENGINEERING & TECHNOLOGY**  
**B.Tech., Summer 2018 - 19 Examination**

**Semester: 8**  
**Subject Code: 03105480**  
**Subject Name: Big Data Analytics**

**Date: 01/05/2019**  
**Time: 10:30 am To 1:00 pm**  
**Total Marks: 60**

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**Instructions:**

1. All questions are compulsory.
2. Figures to the right indicate full marks.
3. Make suitable assumptions wherever necessary.
4. Start new question on new page.

**Q.1 Objective Type Questions (All are compulsory) (Each of one mark)****(15)**

1. Point out the correct statement :
  - a) Hadoop do need specialized hardware to process the data
  - b) Hadoop 2.0 allows live stream processing of real time data
  - c) In Hadoop programming framework output files are divided in to lines or records
  - d) None of the mentioned
2. Hadoop is a framework that works with a variety of related tools. Common cohorts include:
  - a) MapReduce, Hive and HBase
  - b) MapReduce, MySQL and Google Apps
  - c) MapReduce, Hummer and Iguana
  - d) MapReduce, Heron and Trumpet
3. Point out the wrong statement :
  - a) Hadoop processing capabilities are huge and it's real advantage lies in the ability to process terabytes & petabytes of data
  - b) Hadoop uses a programming model called "MapReduce", all the programs should confirms to this model in order to work on Hadoop platform
  - c) The programming model, MapReduce, used by Hadoop is difficult to write and test
  - d) All of the mentioned
4. \_\_\_\_\_ systems are scale-out file-based (HDD) systems moving to more uses of memory in the nodes.
  - a) NoSQL
  - b) NewSQL
  - c) SQL
  - d) All of the mentioned
5. All of the following accurately describe Hadoop, EXCEPT:
  - a) Open source
  - b) Real-time
  - c) Java-based
  - d) Distributed computing approach
6. HDFS provides a command line interface called \_\_\_\_\_ used to interact with HDFS.
  - a) "HDFS Shell"
  - b) "FS Shell"
  - c) "DFS Shell"
  - d) None of the mentioned
7. The daemons associated with the MapReduce phase are \_\_\_\_\_ and task-trackers.
  - a) job-tracker
  - b) map-tracker
  - c) reduce-tracker
  - d) all of the mentioned
8. The \_\_\_\_\_ machine is a single point of failure for an HDFS cluster.
9. Pig operates in mainly how many nodes ?
  - a) Two
  - b) Three
  - c) Four
  - d) Five
10. Hive uses \_\_\_\_\_ for logging.
  - a) logj4
  - b) log4l
  - c) log4i
  - d) log4j

11. Which of the following is not a NoSQL database?
  - a) SQL Server
  - b) MongoDB
  - c) Cassandra
  - d) None of the mentioned
12. Point out the correct statement :
  - a) Hive is not a relational database, but a query engine that supports the parts of SQL specific to querying data
  - b) Hive is a relational database with SQL support
  - c) Pig is a relational database with SQL support
  - d) All of the mentioned
13. Which hdfs command is used to check for various inconsistencies?
  - a) fsk
  - b) fsck
  - c) fetchdt
  - d) none of the mentioned
14. For YARN, the \_\_\_\_\_ Manager UI provides host and port information.
  - a) Data Node
  - b) NameNode
  - c) Resource
  - d) Replication
15. \_\_\_\_\_ command display the list of databases in MongoDB.

**Q.2 Answer the following questions. (Attempt any three)**

**(15)**

- A) Explain horizontal scaling and vertical scaling. Which is used for Hadoop? Justify your answer.
- B) What is the difference between Hadoop and Traditional RDBMS?
- C) Discuss the following terms
  - a. NodeManager
  - b. Job History Server
  - c. Datanode
  - d. Namenode
- D) Discuss fair scheduling in YARN.

**Q.3 A) Introduce Hadoop ecosystem and all of its components.**

**(07)**

- B) Explain Sharding and Replication in NoSQL.

**(08)**

**OR**

- B) Explain anatomy of File Read in Hadoop.

**(08)**

**Q.4 A) Explain five V's of Big data with proper examples.**

**(07)**

**OR**

- A) Define HDFS and YARN, and talk about their respective components.

**(07)**

- B) Explain HIVE architecture

**(08)**

**PARUL UNIVERSITY**  
**FACULTY OF IT & COMPUTER SCIENCE**  
**BCA/B.Sc.(IT)/IMCA(AY\_IV) Summer 2022-23 Examination**

**Semester: 4****Subject Code: (05101266/05102252/05302252)****Subject Name: Big Data Analytics –I****Date: 21/03/2023****Time: 10.30am to 1.00pm****Total Marks: 60****Instructions:**

1. All questions are compulsory.
2. Figures to the right indicate full marks.
3. Make suitable assumptions wherever necessary.
4. Start new question on new page.

**Q.1 Answer the followings.****A. Answer / Define the following in short.****(05)**

1. State the term Big Data.
2. What is structured data?
3. Define: Business Intelligence.
4. Give one disadvantage of Distributed Computing with respect to Big Data.
5. What is the main use of HDFS?

**B. Multiple choice type questions/ Give the sentence true or false. (Each of 01 marks)****(10)**

1. Which of the following is not a characteristics of Big Data?
  - A. Volume
  - B. Variability
  - C. Variety
  - D. Velocity
2. Who among the following do you think would be able to deal with growing number of data source efficiently?
  - A. Business developer
  - B. Data Scientist
  - C. Sales executive
  - D. Web Designer
3. Which application of social network data analysis is used by a customer retention manager?
  - A. Business intelligence
  - B. Marketing
  - C. Product design and development
  - D. Insurance fraud
4. Identify the type of fraud that typically impact online retailers.
  - A. Credit card fraud
  - B. Forward fraud
  - C. Corporate fraud
  - D. Insurance fraud
5. In distributing computing, \_\_\_\_\_.
  - A. The computing task is divided among
  - B. Additional high-capacity disks are added
  - C. The results are shared among several users of the network
  - D. The computing task is moved to the cloud
6. Why are Big Data applications susceptible to latency?
  - A. Big data may reside in a different location from use in-memory
  - B. The volume of big data is too large to be analyzed rapidly
  - C. Big data cannot use in-memory computing
  - D. Big data applications are still in the early stage of development



7. Which of the following is used to denote the small subsets of a large file created by HDFS?
  - A. NameNode
  - B. DataNode
  - C. Blocks
  - D. Namespace
8. What message is generated by a DataNode to indicate its connectivity with NameNode?
  - A. Beep
  - B. Heartbeat
  - C. Analog pulse
  - D. Map
9. Which of the following options most aptly explains the reason behind the creation of Map Reduce?
  - A. Need to increase the processing power of new hardware
  - B. Need to perform complex analysis of structured data
  - C. Need to increase the number of web users
  - D. Need to spared distributed computing
10. In designing the Map Reduce framework, which of the following needs did the engineers consider?
  - A. It should be cheap and distributed free of cost
  - B. Processing should expand and contract automatically
  - C. Processing should be stopped in the case of network failure
  - D. Developers should be able to create new language

**Q.2 Answer the followings. (3 Marks Questions.) (Any Five) (15)**

1. Explain in brief about elements of Big Data.
2. Write any three point of difference between Internal Source and External source of Data.
3. Give brief about the use of social networking in Big Data with example.
4. Explain how distributed computing work with appropriate diagram.
5. Discuss in details the Hadoop Ecosystem.
6. Describe the MapReduce framework with example.

**Q.3 Answer the following. (5 Marks Questions)(Any three) (15)**

1. Explain in detail about evaluation of Big Data with diagram.
2. Discuss with various example, the use of Big Data in preventing Fraudulent activities.
- 3 Describe the Hadoop multi-nodes cluster architecture.
4. Give main significant of NameNode. Explain HDFS architecture in details.

**Q.4 Answer the following in detail.**

- A.** Describe various areas in which decision-making process are influenced by social network data. **(05)**
  - B.(1)** Discuss in details Use of Big Data in Retail Industry. **(05)**
  - B.(2)** Enlist Technologies for Handling Big Data. Give brief about each of them. **(05)**
- OR
- B.(1)** Write steps to setting up and installing Hadoop in its two operating modes. **(05)**
  - B.(2)** Describe with example how to implement file management tasks in Hadoop. **(05)**

Seat No: .....

Enrollment No: .....

**PARUL UNIVERSITY**  
**FACULTY OF IT & COMPUTER SCIENCE**  
**MCA/M.Sc.(IT), Summer 2024-25 Examination**

Semester: 2

Subject Code: 05201295/05202163

Subject Name: Big Data Analytics - I/Introduction to Big Data Analytics

Date: 15-05-2025

Time: 02:00 pm to 04:30 pm

Total Marks: 60

**Instructions:**

1. All questions are compulsory.
2. Make suitable assumptions whenever necessary.
3. Write the answers for both sections on separate answer sheets.

| SECTION - A [30 Marks] |                                                                                                                                                                                                                                                                              |      |     |     |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-----|-----|
| Q1                     | Answer the following questions:                                                                                                                                                                                                                                              | [20] | CO  | BT  |
| (a)                    | Answer <b>all</b> questions                                                                                                                                                                                                                                                  | (5)  |     |     |
| i.                     | What is the purpose of the NameNode in HDFS?<br>(A) To store user data (B) To manage the metadata and namespace of the file system<br>(C) To process data (D) To visualize data trends                                                                                       | 1    | CO3 | BT1 |
| ii.                    | How does virtualization contribute to Big Data environments?<br>(A) It limits resource allocation (B) It provides a way to consolidate hardware resources and improve scalability<br>(C) It reduces the amount of data processed (D) It eliminates the need for any hardware | 1    | CO3 | BT2 |
| iii.                   | What is the importance of 'Velocity' in Big Data?                                                                                                                                                                                                                            | 1    | CO1 | BT1 |
| iv.                    | Name one social networking platform that uses Big Data.                                                                                                                                                                                                                      | 1    | CO2 | BT2 |
| v.                     | Propose a method to optimize data ingestion using Flume for high-throughput scenarios.                                                                                                                                                                                       | 1    | CO3 | BT1 |
|                        |                                                                                                                                                                                                                                                                              |      |     |     |
| (b)                    | Attempt Any <b>Five</b> Questions out of Seven                                                                                                                                                                                                                               | (15) |     |     |
| i.                     | Assess the effectiveness of various virtualization strategies in managing big data workloads.                                                                                                                                                                                | 3    | CO3 | BT4 |
| ii.                    | Design a research project that aims to explore the future trends in Big Data technologies.                                                                                                                                                                                   | 3    | CO1 | BT4 |
| iii.                   | Describe the process by which Big Data can help in identifying patterns of fraudulent transactions.                                                                                                                                                                          | 3    | CO2 | BT3 |
| iv.                    | Illustrate a scenario where a user would need to copy files from the local file system to HDFS using the command line.                                                                                                                                                       | 3    | CO3 | BT3 |

|                               |                                                                                                                                                                                                                                          |             |           |           |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-----------|-----------|
| v.                            | Explain how to implement a big data solution using cloud virtualization.                                                                                                                                                                 | 3           | CO3       | BT3       |
| vi.                           | How do structured, semi-structured, and unstructured data differ in terms of storage and processing requirements?                                                                                                                        | 3           | CO1       | BT3       |
| vii.                          | Propose a framework for integrating Big Data analytics into the insurance claims process to mitigate fraud.                                                                                                                              | 3           | CO2       | BT4       |
|                               |                                                                                                                                                                                                                                          |             |           |           |
| <b>Q2</b>                     | <b>Answer/Solve following in detail (Attempt any 2 out of 3)</b>                                                                                                                                                                         | <b>[10]</b> |           |           |
| (a)                           | Discuss the role of data schema management in the data flow process and how it affects Flume and Sqoop operations.                                                                                                                       | 5           | CO3       | BT4       |
| (b)                           | Compare traditional data processing methods with big data technologies in terms of scalability and performance.                                                                                                                          | 5           | CO3       | BT5       |
| (c)                           | Analyze the advantages and limitations of using HDFS for big data storage compared to traditional file systems.                                                                                                                          | 5           | CO3       | BT5       |
|                               |                                                                                                                                                                                                                                          |             |           |           |
| <b>SECTION - B [30 Marks]</b> |                                                                                                                                                                                                                                          |             |           |           |
| <b>Q1</b>                     | <b>Answer the following questions:</b>                                                                                                                                                                                                   | <b>[20]</b> | <b>CO</b> | <b>BT</b> |
| (a)                           | Answer <b>all</b> questions                                                                                                                                                                                                              | (5)         |           |           |
| i.                            | What is the primary purpose of the awk command in Unix?<br>(A) To display the contents of files<br>(B) To search for specific patterns within text<br>(C) To process and analyze text files in a structured way<br>(D) To compress files | 1           | CO3       | BT1       |
| ii.                           | What is the primary purpose of task execution in a MapReduce job?<br>(A) To visualize the data<br>(B) To process input data and generate output data<br>(C) To store data in HDFS<br>(D) To schedule jobs                                | 1           | CO3       | BT2       |
| iii.                          | Identify the main components of the Hadoop ecosystem and their roles.                                                                                                                                                                    | 1           | CO3       | BT3       |
| iv.                           | Describe the main features of InfoSphere BigInsights.                                                                                                                                                                                    | 1           | CO3       | BT3       |
| v.                            | Which component is responsible for sorting data in Hadoop?                                                                                                                                                                               | 1           | CO3       | BT3       |
|                               |                                                                                                                                                                                                                                          |             |           |           |
| (b)                           | Attempt Any <b>Five</b> Questions out of Seven                                                                                                                                                                                           | (15)        |           |           |
| i.                            | Propose a use case for implementing a Hadoop ecosystem in a retail organization.                                                                                                                                                         | 3           | CO3       | BT4       |
| ii.                           | Describe the differences between public, private, and hybrid clouds in cloud computing.                                                                                                                                                  | 3           | CO3       | BT3       |
| iii.                          | Analyze the advantages and limitations of using Unix tools for data analysis compared to Hadoop.                                                                                                                                         | 3           | CO3       | BT4       |
| iv.                           | Analyze the advantages and disadvantages of different job scheduling algorithms used in Hadoop.                                                                                                                                          | 3           | CO3       | BT4       |
| v.                            | Evaluate the effectiveness of the MapReduce paradigm in the context of modern data analytics.                                                                                                                                            | 3           | CO4       | BT4       |

|           |                                                                                                                                       |             |     |     |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------|-------------|-----|-----|
| vi.       | Propose a MapReduce job that utilizes a combination of different input and output formats to achieve a specific data processing goal. | 3           | CO3 | BT4 |
| vii       | Design a data processing pipeline that integrates IBM Big Data Strategy with Hadoop components for real-time analytics.               | 3           | CO3 | BT4 |
|           |                                                                                                                                       |             |     |     |
| <b>Q2</b> | <b>Answer/Solve following in detail (Attempt any 2 out of 3)</b>                                                                      | <b>[10]</b> |     |     |
| (a)       | Describe how to configure a MapReduce job using the Hadoop command line interface.                                                    | 5           | CO3 | BT5 |
| (b)       | Assess the effectiveness of error handling mechanisms in a MapReduce application and how they can enhance reliability.                | 5           | CO4 | BT5 |
| (c)       | Illustrate a real-world scenario where distributed and parallel computing techniques are used to process large datasets efficiently   | 5           | CO3 | BT5 |
|           |                                                                                                                                       |             |     |     |